

Teichmüller theory Problem set 1

1. For $T \in GL(n, \mathbb{R})$, let

$$M(T) := \sup_{\|v\|=1} \|T(v)\|, m(T) := \inf_{\|v\|=1} \|T(v)\|, K(T) := M(T)/m(T).$$

- a) Show that $K(T) = 1$ if and only if T is a conformal linear map.
- b) Show that $K(T^{-1}) = K(T)$.
- c) Show that $K(ST) \leq K(S)K(T)$.
- d) Show that if A, B are conformal linear maps then $K(ATB) = K(T)$ for all T .
- e) Show that $K(T) = 1$ if and only if $\|T\|^n = |\det T|$.
- f) If $n = 2$, then show that $\|T\|^2 = K(T)|\det T|$.

2. Let $G = SL(2, \mathbb{R})$, $K = SO(2, \mathbb{R})$. For $g \in G$, let $\mu_g \in \mathbb{D}$ denote the Beltrami coefficient of g .

- a) Show that the function $d_0(gK, hK) := \frac{1}{2} \log K(g^{-1}h)$ is a metric on G/K .
- b) Show that the map $G \rightarrow \mathbb{D}, g \mapsto \mu_{g^{-1}}$ induces a bijection $\Phi : G/K \rightarrow \mathbb{D}$.
- c) Define a metric d on $\mathbb{D}$ by $d(\mu, \nu) := d_0(\Phi^{-1}(\mu), \Phi^{-1}(\nu))$, $\mu, \nu \in \mathbb{D}$. For $\mu, \nu \in \mathbb{D}$, show that $d(0, \mu) = d(0, \nu) + d(\nu, \mu)$ if and only if $\nu \in [0, \mu]$.
- d) Let $a_t \in G$ denote the diagonal matrix with entries e^t, e^{-t} . Show that for the metric d on $\mathbb{D}$, the sphere of radius $t > 0$ is given by $S(0, t) = \{ka_t \cdot 0 \mid k \in K\}$.
- e) Show that for any $g \in G$, the map $\phi_g : \mathbb{D} \rightarrow \mathbb{D}, \mu \mapsto g \cdot \mu$ is an isometry of $(\mathbb{D}, d)$, and is a Möbius transformation of $\mathbb{D}$.

3. Let $f : V \subset \mathbb{C} \rightarrow W \subset \mathbb{C}$ be an o.p. C^1 diffeomorphism, let $\mu_f(z) = f_{\bar{z}}/f_z$ be the Beltrami coefficient of f , and let $K_f(z) = K(df_z)$ be the pointwise dilatation of f .

- a) Show that $|\mu_f| = \frac{K_f - 1}{K_f + 1}$.
- b) Show that $d(0, \mu_f) = \frac{1}{2} \log K_f$.

4. Let f, g be K -QC maps defined on $U \subset \mathbb{C}$. Show that $\mu_f = \mu_g$ a.e. if and only if $g = \phi \circ f$ for some conformal map ϕ .

5. Let $0 < R_1 < R_2 < \infty, 0 < R'_1 < R'_2 < \infty$, and let A, A' be the annuli $A = A(R_1, R_2), A' = A(R'_1, R'_2)$.

a) Suppose there is a K -QC map $f : A \rightarrow A'$. Show that

$$\frac{R'_2}{R'_1} \leq \left(\frac{R_2}{R_1} \right)^K.$$

b) Suppose that

$$\frac{R'_2}{R'_1} \leq \left(\frac{R_2}{R_1} \right)^K.$$

Show that there is a K -QC map $f : A \rightarrow A'$.

6) For a path family Γ , let $\lambda(\Gamma)$ denote the extremal length of Γ . Compute $\lambda(\Gamma)$ in the following cases:

- a) Γ = curves joining b-sides of a rectangle with sides a, b .

b) Γ = curves joining boundary components of an annulus $A(R_1, R_2)$ where $0 < R_1 < R_2 < \infty$.

c) Γ = Jordan curves separating boundary components of an annulus $A(R_1, R_2)$ where $0 < R_1 < R_2 < \infty$.

7 Show that if Γ_1, Γ_2 are path families such that $\Gamma_1 \subset \Gamma_2$ then $\lambda(\Gamma_2) \leq \lambda(\Gamma_1)$.

8 Let $A_1, A_2 \subset \mathbb{C}$ be annuli. Show that if $A_1 \subset A_2$ then $m(A_1) \leq m(A_2)$.

9 Let $f : U \subset \mathbb{C} \rightarrow V \subset \mathbb{C}$ be a C^1 K -QC map. Show that for any path family Γ in U , we have $\lambda(f(\Gamma)) \leq K\lambda(\Gamma)$.

10 Let X, Y be Riemann surfaces whose universal cover is $\mathbb{H}$, and suppose there is an o.p. homeomorphism $f : X \rightarrow Y$. Show that f is K -QC if and only if any lift $\tilde{f} : \mathbb{H} \rightarrow \mathbb{H}$ of f is K -QC.

11 Show that there does not exist a QC map $f : \mathbb{C}^* \rightarrow \mathbb{D}^*$.

12 Show that there does not exist a QC map $f : \mathbb{C} \rightarrow \mathbb{D}$.

13 Show that if there is a QC map $f : X \rightarrow Y$ between Riemann surfaces X, Y , then the universal covers of X, Y are biholomorphic.

14 Let X be a Riemann surface, and suppose there is a K -QC map $f : X \rightarrow \mathbb{D}^*$. Show that there is a biholomorphic map $g : X \rightarrow \mathbb{D}^*$.

15 Let $D \subset \mathbb{C}$ be an open set and let $\mu \in L^\infty(D)$ be such that $\|\mu\|_\infty = k < 1$. Let $\{(U_i, \phi_i : U_i \rightarrow \mathbb{C})\}_{i \in I}$ be a collection of pairs such that $\{U_i\}_{i \in I}$ is an open cover of D , and ϕ_i is a QC-map on U_i such that $\mu_{\phi_i} = \mu$ a.e. on U_i , for all $i \in I$. Show that $X = (D, \{(U_i, \phi_i)\}_{i \in I})$ is a Riemann surface, and the identity map of D viewed as a map $id_D : D \subset \mathbb{C} \rightarrow X$ is a K -QC map, where $K = (1+k)/(1-k)$.

16 Let $f_n : \mathbb{C} \rightarrow \mathbb{C}$ be a sequence of K -QC maps such that $f_n(0) = 0, f_n(1) = 1$ for all n , suppose $\mu_{f_n} \rightarrow 0$ a.e. on $\mathbb{C}$, and suppose $f_n \rightarrow f$ uniformly on compacts in $\mathbb{C}$ for some o.p. homeomorphism $f : \mathbb{C} \rightarrow \mathbb{C}$. Show that $f = id$.